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Neuroelectrophysiological correlates of extended cessation of consciousness in advanced meditators: A multimodal EEG and MEG study

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doi: https://doi.org/10.1101/2025.09.19.677455

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Abstract

In some contemplative traditions, 'extended cessation' (EC) refers to a state of advanced meditation in which the meditator self-reports intentionally suppressing their own consciousness and re-emerging with a profound sense of clarity and equanimity. Here, we present the first electrophysiological study of EC, in which five meditators underwent concurrent electroencephalography (EEG) and magnetoencephalography (MEG) recording. EC significantly increased high-frequency global coherence in all five of the participants, yet it did not consistently affect Lempel-Ziv complexity or power across participants. Our results provide preliminary evidence that the neural signatures of EC may be distinct from other states that are characterized by a loss of consciousness, including deep sleep and anesthesia. EC may not be simply 'switching off' consciousness but rather reorganizing the neural dynamics that underlie consciousness.

Competing Interest Statement

The authors have declared no competing interest.

FUNDER INFORMATION DECLARED

Dimension Giving Fund

Tan Teo Charitable Foundation

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bioRxiv and medRxiv thank the following for their generous financial support:

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Posted October 17, 2026.

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